

Technology Monitoring Report

Theme: Artificial Intelligence in IT and Software Development

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Course: Technology Monitoring & English Presentation | Date: June 2026

Summary

1. Our monitoring strategy
2. Monitoring feedback (synthesis): Recent developments of AI in IT
3. Specialized article: GitHub Copilot and LLMs in Software Development

1. Our Monitoring Strategy

Professional interests

General	IT support, network administration, systems security, and web/application development.
Specialized	Artificial Intelligence (AI) tools, Large Language Models (LLMs) for code generation, prompt engineering, and automation in IT administration.

List of professional monitoring sites

- **General Professional Sites:** TechCrunch.com, Wired.com, Developpez.com, Le Monde Informatique, and YouTube (@Micode).
- **Specialized Professional Sites:** OpenAI Blog (openai.com/blog), GitHub Blog (github.blog), Hugging Face News (huggingface.co), InfoQ AI section, and YouTube channels (@Fireship, @TechWithTim).

Selected tool

Feedly: We chose Feedly as our main RSS aggregator tool. It allows our group of 4 students to organize feeds into categories (like "AI Tools" and "Cybersecurity") and follow updates every day from one interface.

Domain references

- **Yann LeCun** (Chief AI Scientist at Meta) — twitter.com/ylecun
- **Andrew Ng** (Founder of DeepLearning.AI and Stanford Professor) — twitter.com/AndrewYNg
- **Andre Karpathy** (AI Developer and former Director of AI at Tesla) — github.com/karpathy

Our communication style

We work as active readers. We collect articles every week, verify the facts by comparing different sources, and write short reports. We share our summaries with our classmates during oral presentations and via our group's shared document tool.

2. Monitoring Feedback (Synthesis): Recent Developments of AI in IT

Artificial Intelligence continues to grow very fast. In recent months, generative AI has become a mandatory assistant for IT workers rather than just a curiosity. Here is an overview of the most important trends and their concrete interest for our future IT careers.

AI Assistants are the New Standard for Coding

New versions of large models (like GPT-4o, Claude 3.5, and specialized code models) are now perfectly integrated into code editors like Visual Studio Code. These tools do not just write simple snippets anymore; they can analyze an entire workspace, find security vulnerabilities, and generate automated tests. For developer teams, this reduces time spent on repetitive tasks and helps focus on application logic.

AI in System Administration and Networks

For network and system technicians, AI helps to automate infrastructure monitoring. Modern IT tools use machine learning to scan server logs and identify strange activities or security threats immediately. This allows administrators to prevent server crashes and block cyberattacks before they damage the network ecosystem.

Operational Interest for BTS SIO Students

As first-year students, learning "Prompt Engineering" (how to talk to an AI correctly) is becoming essential. We need to know how to give precise instructions to obtain good code or scripts. However, human control is still necessary because AI can invent wrong answers (hallucinations). We must always test and understand the code before using it.

3. Specialized Article: GitHub Copilot and LLMs in Software Development

A Powerful Duo for Developers

GitHub Copilot is an AI-powered code assistant developed by GitHub and OpenAI. It works directly inside your Integrated Development Environment (IDE). By combining human creativity with the power of Large Language Models (LLMs), it creates a very efficient workspace for modern programmers.

Operational Synergy

Next-generation programming relies heavily on cooperation between the developer and the AI assistant. For example, when you write a comment like `// Function to verify an email with a regex`, GitHub Copilot automatically suggests the full block of code in JavaScript, Python, or PHP. This synergy allows developers to write code much faster. It also helps students learn new programming languages by showing real-time examples of code structure and syntax rules.

Tips for Getting Started with AI Code Assistants

- 1. Install the proper extensions:** Add GitHub Copilot or equivalent free tools (like Codeium or Ollama for local execution) to your IDE.
- 2. Write clear comments:** The AI works better if your comments explain the context and the exact input/output variables.
- 3. Keep critical thinking:** Never accept a code suggestion blindly. Review the syntax, test the performance, and ensure it respects security rules. You are the pilot, the AI is only the copilot!

Conclusion

The combination of AI tools and standard IT skills offers a solid foundation for building modern software and secure networks. By understanding how to control these technologies today, our group is preparing for the real needs of the IT job market.